Control and Locomotion of Tensegrity Robots through Manipulation of the Center of Mass

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***Abstract* - This paper provides the methodology used to simulate and control an icosahedral tensegrity structure augmented with moveable masses attached to each bar which provides a means of locomotion. The center of mass of the system can be changed by moving the masses along the length of each of the bars that compose the structure. Moving the masses changes the moments created by gravitational force, allowing for the structure to roll. With this methodology in mind, a controller was created to move the masses to the desired locations to cause such a roll. As shown later in this paper, such a methodology, assuming the movable masses have the required mass, allows for full control of the system using a quasi-static controller created specifically for this system. This system has advantages over traditional tensegrity controllers because it retains it's shape, and is designed for high shock scenarios in mind.**

I. INTRODUCTION

Tensegrity systems refer to a class of pin-jointed structures which consist of contiguous cables in tension and non-contiguous bars in compression. The tension in the cables stabilize the structure in a mass efficient way, since tensegrity structures are capable of minimizing mass needed for buckling loads. With pretensioned cables, many designs are statically stable. Since the mid twentieth century, tensegrity structures have become popular among artist and engineers alike [1]. Among the engineering community, such structures sparked a broad interest due to their unique nature of combining shock absorption, minimizing the mass of a system, especially in buckling conditions and the capability to collapse to small volumes [2,3,4,5].

More recently, select groups have studied these structures for various robotic applications [6, 7], especially to see the feasibility of such structures acting as a lunar landing robot.[8] These applications, in which using a tensegrity structure brings the significant advantages of large amounts of shock absorption, the capability to collapse, and low mass to stress ratios compared to other solutions, make it necessary to create a reliable, simple control system for these structures.

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One solution for controlling a tensegrity structure is to use gravity as the exclusive means of control. The only thing needed for such a structure to roll is for the center of mass of the structure to be outside of the current structure's support polygon. When this occurs, the structure will rotate towards the location of the center of mass. So, to control the rotation of the structure, and thereby the location of the structure, it is only necessary to control the location of the center of mass of the structure.

Past researchers have addressed this problem by controlling the tension of the cables and thereby the shape of the structure [8,9,10,11]. Changing the length and tension on each string causes the structure to change its shape. If the shape changes drastically enough, the structure will roll. However, this method involves attaching large, bulky actuators to the structure to control each of the cables. When actuators are attached to the end of each bar, there is a greater chance for damage upon impact with the ground. The dynamics presented by changing the cable lengths of the system are also more complicated and computationally expensive. The model proposed here would have actuators for the masses located within the bar, allowing for optimal actuator placement to mitigate damage from impact conditions. These actuators would also be small enough to fit within each bar. Future work could include using magnets to move the masses, removing the need for actuators entirely.

Rather than changing string tension and lengths, this work focuses on using moveable masses attached to each bar of the system to adjust the center of mass and thus induce motion. While using movable masses as a means of control has been explored previously in other applications [12], it has not yet been applied to tensegrity structure control.

This paper introduces an approach that implements a simple control method which allows the robot to control itself under a quasi-static assumption. The controller itself activates when the system is at rest. When at rest, the controller choses the node – one of the vertexes of the system's physical shape- that the system desires to roll to, and then moves the masses to the calculated locations. Such an action moves the system's center of mass outside of the support polygon in the desired direction of movement, allowing for the system to move.

This method allows for only 6 unique variables to be used for control, one variable for each of the movable masses, instead of several more variables needed when controlling the cables of the system. Cable driven systems have up to 24 cables, several of which need to be altered at once to get the system to move correctly. It does not involve bulky actuators at the ends of the rods, rather the actuators can be part of the moveable mass. Additionally, the system is not required to change any of its inherent properties (such a string length or tension), thus allowing it to retain its initial shape at all times and allows the system to be over constrained for redundancy. By retaining its shape, the control of the system is very predictable over all orientations of the system, since moving the masses to control the system is very similar in every system orientation.

Ultimately, this paper hopes to contribute a method of control based solely on mass actuation across the bars of the model, with no alterations by the system on the length of the cables of the system. It will show the feasibility and the simplicity of such a system in flat conditions.

The remainder of the paper contains the following sections: Section 2 introduces the environment that was created for the model, including how the movement of masses was simulated. Section 3 explores mathematically the feasibility of the moveable mass method as well as the constraints on the system in order to guarantee movement. Section 4 introduces the controller and explains the necessary assumptions. Section 5 contains a discussion of the results generated by the model as well as a comparison with a physical system. Finally, Section 6 summarizes the current findings and discusses future work to be done.

II. MODELING THE TENSEGRITY ENVIRONMENT

The tensegrity model developed for this work builds on the standard dynamic tensegrity model developed in MATLAB by MOTES [13], with the same traditional assumptions that quasi-static tensegrity structures use. This model, whose shape was chosen for its convenient properties of the bars creating three orthonormal planes, as shown in Fig. 1, was translated into Python from the MATLAB code mentioned above and then modified to fit the needs of the robotic system. These modifications are explained below.

The first major change that was made to the code was to remove the requirement for the system to have a pinned node. In order to allow the system to roll around, no nodes can be permanently pinned to ground. The pinned nodes to ground were replaced with a spring and damper used to mimic the physical interaction with the ground. This solution allows for the model to roll without constraining the bottom of the model with a no velocity constraint, which can cause unwanted energy to be added to the system, thus permitting it to react to the ground as the real system would, without the unpredictable consequences of trying to make the ground a pinned node.

The second change was to model moveable masses that could move across the bars. To do this, forces were added to either node of each bar. Each mass could then be moved by changing the forces on each node of the bar, where the total force from the two nodes combined remains constant. Thus, in this model, the masses used to control the bar are simply modeled by the gravitational forces that the masses would emit.

This force approach can be used instead of modeling masses for two main reasons. First, since the system is quasi static for the current modeling approach, the fidelity of the dynamics does not have to be particularly high. This assumes that the system has settled before the next controller iteration. The mass can be neglected, which greatly simplifies the dynamics of the model, while still reasonably simulating the roll characteristics for this system.

Second, to reduce excess movement, a larger damping term is used. Again, the fidelity of the dynamics can be reduced to simplify the model. Thus, by modeling the ground as a spring and the movable masses as two forces on either side of each mass' corresponding bar, it was possible to create a model that works well with quasi-static control. If a quasi-static control is not used, the model would have to be evaluated to see if it would still work within whatever error tolerance was expected.

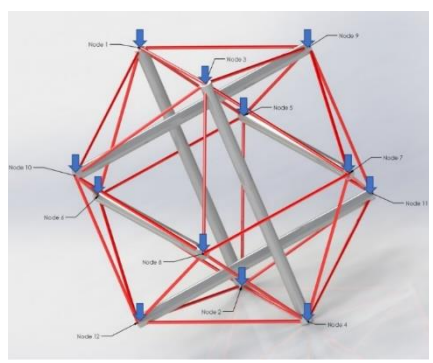

Figure 1: The above figure shows the nominal configuration of the tensegrity structure. The bars (in grey) are where the moveable masses reside.

III. SYSTEM ROLL CONDITIONS

The tensegrity system chosen for control is an icosahedron, or a 20 faced structure where each face is a triangle. This shape is composed of 12 vertices (nodes), with each node connected to the five closest nodes. Each of these nodes are located according to a permutation of the following coordinates: $(0, \pm 1, \pm\phi)$, $(\pm 1, \pm\phi, 0)$, $(\pm\phi, 0, \pm 1)$, where ϕ is the golden ratio.

With this structure, it is possible to statically model the system as if gravitational force of the moveable masses is split equally between the two nodes at the end of each bar. This is a valid assumption for the dynamics of the model since a quasi-static controller is used.

The center of mass of the system can be modeled as

$$x_{cm}^{n} = \frac{1}{m_{tot}} \sum_{i=1}^{12} m_i x_i^n \tag{1}$$

where n is the $\hat{\imath}, \hat{\jmath}, and\ \hat{k}$ respectively.

Assuming each of the nodes carry the same mass, (1) can simplify to:

$$x_{cm}^{n} = \sum_{i=1}^{12} x_i^n \tag{2}$$

Solving (2) reveals that the center of mass of the system is located at (0,0,0) in its initial condition. For example, in Fig. 2 below, the origin of the initial coordinate frame would be located in the center of the structure, at its center of mass. In this paper, the xz plane will be parallel to the ground.

In any resting state, the system rests on a triangular support polygon. Due to the geometry of the regular icosahedron, each of these triangles are equilateral triangles. Due to the symmetry of the system, the center of mass of the system is directly above the center of the support polygon in any configuration.

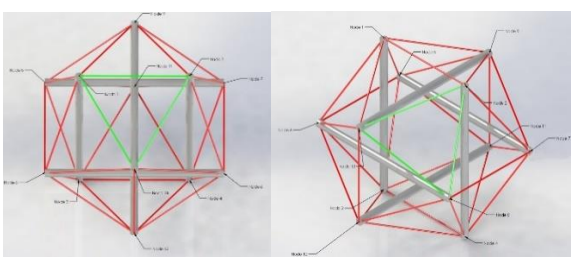

Figure 2: Viewed from below, the left image shows the support polygon when the structure in its spiral configuration. The right image shows the support polygon when the structure is in its parallel configuration. The green triangle represents the system's support polygon in each configuration.

There are two main configurations that describe the state of this tensegrity system, spiral and parallel. A visualization of those cases is shown in Fig. 2 above. In both cases, the way to induce rolling in the structure is to move the center of mass of the entire system to outside of the support polygon. To do this, the moveable masses must overcome the system's own inertia. To determine the required mass, first find the distance from the center of mass of the system to the edge of the support polygon triangle around which it will be rotated. Next, find the 2D equation to represent the vector to be rotated about. The 2D equation for this vector is created by finding the equation of the line between two of the nodes of the support polygon and neglecting the vertical component of the line. Once this vector is obtained, it is possible to find the moments from the various masses of the system by finding length of the segment perpendicular to the rotation vector and any of the nodes selected to rotate about.

Using the standard moment equation, the summation of moments for the system is:

$$\sum M = \left(\sum_{i=1}^{6} r_i\, X(-m_m g\hat{\jmath}) \right) + r_c\, X(-m_c g\hat{\jmath}) \tag{3}$$

here r_i is the moment arm for each respective movable mass and r_c is the moment arm for the center of mass, while m_m is the mass of each moveable mass and m_c is the mass of the tensegrity structure.

In the model, there are two different configurations of the tensegrity system as shown in Fig. 2. In the two configurations, the possible placements of the movable masses change the moment arms that can be used.

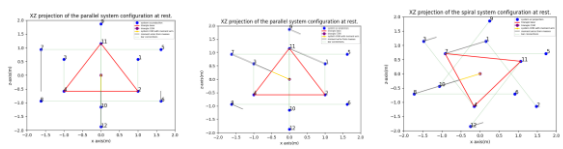

Figure 3: The figure shows the possible orientations of the system, and the moment arms that can be applied on that system. In the spiral orientation (rightmost image), all of the possible directions have the same moment arms, while in the parallel orientation (left and middle image) there are two potential configurations for moment arms.

In each scenario, the sum of moments can be determined by summing the moveable mass moment arms and then multiplying by each respective mass. The moment of the center of mass is also included in the following:

$$\sum M = \left(\sum r_m \right) m_m - (r_c) m_c \tag{4}$$

Solving numerically for each scenario, where each scenario is shown in Fig. 3 above, yields the following results:

$$m_m > 0.8089 m_c \tag{5}$$

$$m_m > 0.4058 m_c \tag{6}$$

$$m_m > 0.4045 m_c \tag{7}$$

These results show what proportion of the total mass of the system the movable masses have to be to get the system to actuate.

An interesting result due to this study is that the system can get onto any support polygon with each movable mass being only 40% of the original system mass. This is due to rotation requiring the mass outlined in (5) can be avoided.

IV. CONTROLLER IMPLEMENTAION

The quasi-static controller for this system relies on a few key questions that will be discussed in the following section. First, given a valid desired node, meaning a node that is initially connected to two support polygon nodes, but is not part of the support polygon, how can the masses be commanded such that the system rolls to that node? Second, how is the commanded node chosen? - meaning if the model is commanded to move in a specific direction (up, down, left, right), how is the node chosen to maximize the system's movement in that direction.? Finally, how does the controller determine what direction to move to reach a desired target?.

Each node in this icosahedral tensegrity system is connected to five other nodes via cables, as illustrated in Fig. 4. Two of these nodes are on the same bar, but each of the other nodes are connected to unique bars. Including the node that the model desires to move towards, the cable connections inform the model where to move four of the six masses. In Fig. 5, the figure on the left shows the system in a configuration where the masses are in the center of each bar, with the masses represented by the blue circles. The distribution of the forces created by the masses on each respective bar is shown with the blue arrows. The target node is shown as the green circle in the image and the desired location of the other masses for movement are shown as a yellow circle. The middle image shown how the forces on each node change when the masses are moved, and the image on the right shows what the system looks like after the masses have moved. For clarity, the relevant masses should move entirely to either the desired node or to one of the direct connections. This makes it so the system has the maximum moment around the edge of the support polygon so the system rotates around the correct pivot. The location of the other masses – the mass on the bar not represented and the mass on the bar represented twice, which is shown in Fig. 4 – matters only if fine control of the system is desired. This configuration would also allow for the system to accelerate the mass down the length of the bar, as discussed in section 3.

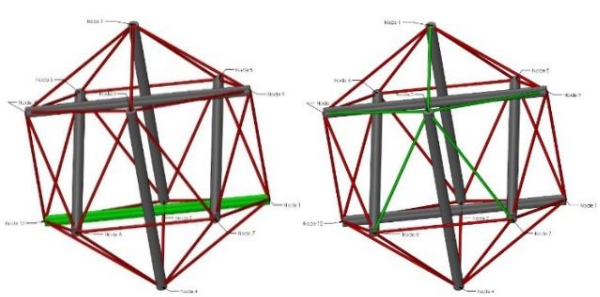

Figure 4 The images shows a node connection. The image on the left highlights the unconnected bar, while the image on the right shows the string connections coming off the desired node.

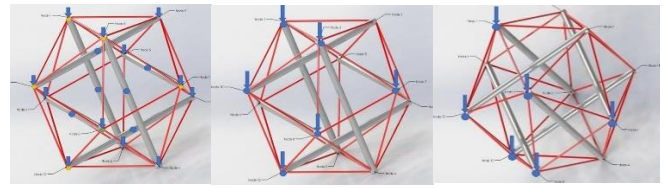

Figure 5: The figure shows the process used to move the masses to the location required to make the system roll to a desired node.

With that in mind, if the system is controlled to the worst case scenario represented by (5), then the placement of the masses only matters if the user wants to prevent momentum rolls, where the system does not stop at the next triangular polygon support and instead moves past it to a new orientation. If this is the case, the masses should go to the node that has the smaller moment arm. This will act as a damper for the system, slowing the initial roll and preventing an orientation that would cause the model to roll again once it has initially settled. Over rolling happens quite readily in practice, since a combination of the momentum from the system changing orientation and the locations of the masses essentially primes the system to continue to roll. Damping with the mass is a technique to prevent that behavior.

Now that the system can move towards a desired node, the next step is to algorithmically determine where the system should move. Consider the positive x direction as an example, where the x direction follows the x direction as described in Fig. 3. The support polygon of the icosahedral structure is composed of a triangle of cables connecting three nodes at y = 0. This is the current support polygon of the system. Each support polygon node has its own x and z locations, each of which are unique to that node at that system state. Each of the string connections are vectors about which rotation is possible. Thus, at any particular step there are three possible vectors for the system to rotate about.

To narrow down the options for which way to rotate the system, the following process is followed. The first step is to choose the node with the largest value in the direction chosen to rotate in. In the positive x case, it would be the node with the largest positive x value. Choosing this node disqualifies the vector between the two other nodes as the rotation vector, since rotation about that vector would cause the system to go in the direction opposite of the desired location.

Once that node is established, the vectors between the chosen node and the other two nodes are created. The vector that is most perpendicular to the desired direction is then chosen, since rotation about that direction leads the system to change its orientation in that direction the most. This determination is done by choosing the minimal absolute value of the dot product between the direction vector, [1,0,0] for positive x, and the calculated vectors between the first node chosen in the support polygon and the other potential nodes that remain in the triangular support polygon.

Finally, to evaluate the controller to move the system to a desired target, the following methodology is used. First, a random target in xz space is created for the model to move to. The controller has functions to move the system in both the positive and negative x and z directions. So to choose the direction that the system needs to go, the algorithm finds which direction the system needs to move the most, and then moves it in that direction.

The choice of direction is made using a greedy algorithm that finds the minimum distance from each node to the target in the positive and negative x and z directions. Once those minimum distances are found, it chooses the largest one and has the system roll in that direction. If all the distances are negative (meaning that it's support polygon contains the required target), then the system made it to the target, which is the success criteria of the algorithm. If further simulation is required, then another random point is chosen and the simulation continues. The minimum negative z and positive z are not simply the negation of the other because it is the minimum of several nodes, thus necessitating that those values be different. Other controllers have been created for tensegrity systems that work on collision avoidance and other path planning optimizations. Such techniques could be used to refine this controller in the future [14,15].

V. RESULTS

In this section the responses of the simulated and real systems are obtained and analyzed. Simulation testing was performed to see if the system can control to its specified targets and to see how the dynamics of the system work over many control iterations.

The first test performed in the simulated system was to rotate the system towards a desired node. To do so, the system was put into its initial configuration and one of the 3 possible nodes were chosen as the target node. The system was then commanded to move to that particular node and it rolled to that node using the methodology described in section 4. Fig. 7 shows two specific tests. On the left, the system was commanded to go up twice, left twice, down twice, and right twice. Due to the triangular shape of the polygonal support, the system travels in only a rough square shape. On the right, the system is commanded to go to a random target location. Using the procedure also outlined in section 4, the system was able to move in as directed and was able to transverse to the random target.

To ensure the feasibility of this moveable mass model, a real-world structure was created using aluminum pipes and large removable masses. These masses could be hung on either end of each bar to verify that moving these masses could make the system roll. The required mass of the large masses was initially calculated using the equations derived in section 3 of this paper.

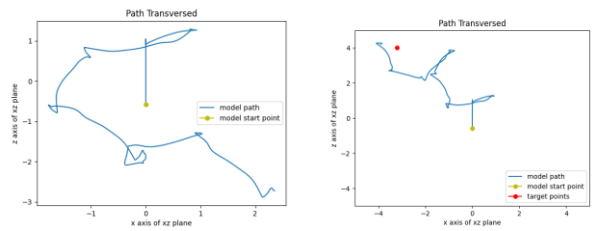

Figure 7: The figures shows the path of the model. The image on the left is when the system is commanded to go up twice, left twice, down twice, and right twice. The right image is when the system is commanded to go to a random target using the same control method.

After finding the mass of the initial system, the required mass was calculated for each unique orientation of the system, and that mass was initially placed on the relevant nodes of the system to induce rotation. The masses were then adjusted to experimentally find the minimum mass needed to get the system to rotate.

From experimentation, the actual values of the masses needed for the three different roll configurations are the following:

Parallel planes configuration, worst case:

$$m_m >* m_c \tag{8}$$

Where * means rotation was not achieved.

Parallel planes configuration, best case:

$$m_m > (.208)m_c \tag{9}$$

Spiral configuration:

$$m_m > (.338)m_c \tag{10}$$

Fig, 6 shows the structure before and after rotation. As can be seen from this analysis, the predicted mass-to-system mass comparison the spiral configuration and the parallel planes best case configuration both overachieved the predictions from the paper above. This could be due to the size of the base differing from that which was used in the analytical calculations, as the actual structure is not a perfect icosahedron.

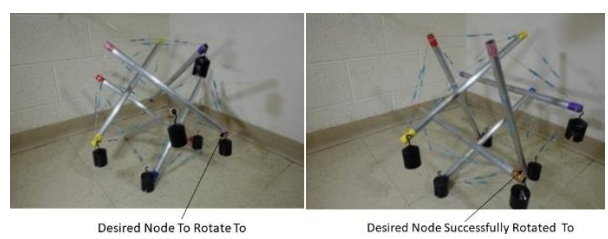

Figure 6: The structure is shown on its spiral (equilateral) support polygon, loaded to rotate onto a parallel planes (isosceles) support polygon. 1kg

weights are used for simplicity and to reduce visual clutter in this picture. In this illustration, the pink node at the bottom right of the image is the node to be rotated to. The left image is prior to roll, while the right image is after roll.

However, due to this change of state, this means that the worst-case scenario is even worse for rolling. The isosceles support polygon essentially moves the nominal system center of mass away from the rolling axis, which would make the system harder to roll. The bars also are closer together than would normally be described, thus reducing moment arms from the bars in this case.

VI. SUMMARY AND FINAL REMARKS

Traditional approaches for controlling tensegrity structures typically require the actuation of the tensegrity tensile members to allow for the movement of the system. This approach leads to complicated dynamics, since there are 32 design parameters as described in [11]. This method circumvents that challenge by using only 6 movable masses within the bars of the system as the means of locomotion, which reduces the number of bulky actuators, and makes the system more shock resistant due to its less exposed critical equipment.

Movable masses are a valid strategy as a means of system locomotion without the masses of the movable masses being significantly higher than that of the system.

In essence, this model is capable of causing the system to rotate, move in a particular direction, and control to a point in space. It is capable of achieving this with computationally cheap control and simple algorithms.

This simple approach to the system allows for less moving parts and for the most sensitive parts of the system to be protected by the bars. This would allow for the system's most critical components to be protected from shock, allowing the system to operate under the conditions that it was created to cope with.

Future work will include making a physical system that is capable of moving the masses in the requisite manner to allow for control. This involves designing the movable mass – bar system. Additionally, a new controller will have to be designed to work with the limited inputs that the physical system will have. Another practicality that this controller will need to be able to deal with is that in practice, there are six edges along which the system will not be able to rotate. Additional work could include mass momentum methods to allow the system to rotate in a more controlled manner when desired for more optimal control. In addition, this type of system would be a good candidate to use reinforcement learning for path planning and actuation of masses since such an approach would treat the terrain as a maze and it would be possible to learn on such a dataset.[16]

Once the physical system has been established, along with the ability to control that system, then optimizing the system to be impact resistant would be the next step of design, which would include location of masses and payloads to ensure maximum likelihood of system survivability.

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